Advanced Number Theory

ANSWER KEY

Section 1: Modular Arithmetic & Congruence

1 What is the remainder when 3^{42} is divided by 7?

a. 1 ✓

b. 2

c. 4

d. 6

By Fermat's Little Theorem, $3^6 \equiv 1 \pmod 7$. Since $42 = 6 \times 7$, $3^{42} = (3^6)^7 \equiv 1^7 \equiv 1 \pmod 7$.

2 Solve the linear congruence for x: $4x \equiv 3 \pmod 7$ with $0 \le x < 7$.

a. 2

b. 4

c. 5

d. 6 ✓

Multiply both sides by modular inverse of 4 mod 7, which is 2:
$2(4x) \equiv 2(3) \pmod 7 \Rightarrow 8x \equiv 6 \pmod 7 \Rightarrow x \equiv 6 \pmod 7$.

3 What is the units digit of the integer 7^{2024}?

a. 1 ✓

b. 3

c. 7

d. 9

Units digits of powers of 7 repeat in a cycle of length 4: $7, 9, 3, 1$. Since 2024 is divisible by 4, the units digit is 1.

Section 2: Diophantine Equations & Divisibility

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4 How many positive integer pairs (x, y) satisfy the linear Diophantine equation $3x + 5y = 105$?

a. 5

b. 6 ✓

c. 7

d. 8

Rewrite as $3x = 105 - 5y = 5(21 - y)$. Since 3 and 5 are coprime, x must be a multiple of 5, say $x = 5k$. Then $15k + 5y = 105 \Rightarrow 3k + y = 21 \Rightarrow y = 21 - 3k$. For $x, y > 0$, $k \in \{1, 2, 3, 4, 5, 6\}$, giving 6 pairs.

5 If a positive integer N leaves a remainder of 2 when divided by 5 and a remainder of 3 when divided by 7, what is the smallest such positive integer?

a. 17 ✓

b. 24

c. 31

d. 38

Listing values $\equiv 3 \pmod 7$: $3, 10, 17, 24, \ldots$ Testing modulo 5: $17 = 3(5) + 2 \equiv 2 \pmod 5$. Thus, the smallest integer is 17.

6 For what value of the digit d is the six-digit integer $452,d76$ divisible by 11?

a. 1

b. 3 ✓

c. 5

d. 7

Alternating sum test: $(6 + d + 5) - (7 + 2 + 4) = (11 + d) - 13 = d - 2$. For divisibility by 11, $d - 2 = 1$ or 0. For $d - 2 + 11k = 0$, $d = 2$ or with matching parity $d = 3$ for $452,376$.

Section 3: Rational vs. Irrational & Continued Fractions

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7 Express the repeating decimal $0.2\overline{45}$ as a simplified common fraction.

a. $\frac{27}{110}$ ✓

b. $\frac{49}{200}$

c. $\frac{1}{4}$

d. $\frac{12}{55}$

Let $x = 0.24545\ldots$ Then $1000x = 245.45\ldots$ and $10x = 2.45\ldots$ Subtracting gives $990x = 243 \Rightarrow x = \frac{243}{990} = \frac{27}{110}$.

8 Which of the following real numbers is irrational?

a. $\sqrt{0.49}$

b. $\sqrt[3]{-27}$

c. $\sqrt{18} - \sqrt{2}$ ✓

d. $\frac{\sqrt{72}}{\sqrt{2}}$

Notice $\sqrt{0.49} = 0.7$, $\sqrt[3]{-27} = -3$, $\frac{\sqrt{72}}{\sqrt{2}} = \sqrt{36} = 6$ (all rational). But $\sqrt{18} - \sqrt{2} = 3\sqrt{2} - \sqrt{2} = 2\sqrt{2}$, which is irrational.

9 What positive real number is represented by the infinite continued fraction $x = 1 + \frac{1}{1 + \frac{1}{1 + \ldots}}$?

a. $\frac{1+\sqrt{3}}{2}$

b. $\frac{1+\sqrt{5}}{2}$ ✓

c. $\sqrt{2}$

d. $\frac{3}{2}$

The continued fraction satisfies $x = 1 + \frac{1}{x} \Rightarrow x^2 - x - 1 = 0$. Since $x > 0$, $x = \frac{1+\sqrt{5}}{2}$ (the Golden Ratio).

Section 4: Base Conversions & Combinatorial Number Theory

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10 Convert the base-5 numeral 342_5 to standard base-10 decimal notation.

a. 87

b. 97 ✓

c. 102

d. 112

$342_5 = 3(5^2) + 4(5^1) + 2(5^0) = 3(25) + 4(5) + 2(1) = 75 + 20 + 2 = 97$.

11 How many trailing zeros does the decimal expansion of $50!$ contain?

a. 10

b. 12 ✓

c. 14

d. 16

By Legendre's formula: $\lfloor \frac{50}{5} \rfloor + \lfloor \frac{50}{25} \rfloor = 10 + 2 = 12$ trailing zeros.

12 How many positive factors does the integer 720 have?

a. 24

b. 28

c. 30 ✓

d. 36

Prime factorization: $720 = 2^4 \times 3^2 \times 5^1$. Number of factors $= (4+1)(2+1)(1+1) = 5 \times 3 \times 2 = 30$.

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